OpenSourceRail v0.5

Ferrite PM maglev — primary design reader edition

Fixed magnetic modules · compact elevated guidebeams · Rust controls · local manufacture

Provisional digital studies. Complete vehicle, civil and software qualification remains open. Accepted physical releases: 0.

OpenSourceRail v0.5 — ferrite PM maglev

docs/open-source-rail-overview.md

Fewer moving parts, local manufacture and compact elevated urban routes.



OpenSourceRail maglev station concept

The primary design uses fixed onboard ferrite magnetic modules and powered linear electromagnetic propulsion, replacing conventional wheelsets, axles, gearboxes and rotating traction motors in normal running gear. Windings, drives, cooling, sensors, protection, passenger doors and emergency systems remain necessary. Contactless running aims to reduce wheel/rail contact noise and wear; acoustic performance and whole-system maintenance still require measurement.

Simple civil works and local capability

The research target is one compact integrated concrete guidebeam per direction, with necessary ferromagnetic reaction surfaces, repeatable supports and no continuous external walkway. A vertical spiral evacuation chute is the preferred alternative under investigation. Walkway removal has not been approved. Route clearance, safe landings, assisted evacuation, stopping/recovery and fire response must work at every permitted stopping location.

The 99% elevated-mainline objective and straight tangents aim to reach constrained urban corridors with less ground occupation and disruption. Land rights, utilities, access, wildlife, waterways, curvature and foundation feasibility still govern each route. No universal clearance or zero-disruption claim is made.

Local casting, steel fabrication, cartridge assembly, winding, vehicle structures, interiors, wiring, Rust software and integration can create domestic capability. The complete local-content percentage, installed programme cost and funding requirement remain unknown until origin-traced bills, processes and supplier offers exist. Earlier LM3 factory-gate prices and portfolio percentages are retained comparison inputs, not estimates for the new system.

Inspect and reproduce the current work

Current digital capability	Scope
City planning	266 separate primary maglev packages; retained geography/demand, recalculated formation, duty, fleet and charging
Magnetic interface	624 native solids; partial lift/cartridge/frame geometry
Beam-only civil kit	54 native solids; selected geometry without external walkways
Evacuation entry and support	20 partial entry parts and 3 support envelopes; not complete manufactured products
Controls	Allocation-free Rust core, 24-module replay, I/O contracts and advisory evacuation state machine
Simulation and media	Field equation controls, actual two-way Rust/beam replay, four assembly visibility videos, native images and source receipts
Engineering acceptance	0 accepted physical releases; design, manufacture and operational gates remain open

Use `./osr build` to verify dependencies and rebuild stale implemented studies. `./osr workbench` opens the primary maglev UI. `./osr design check` independently regenerates the report. `./osr book` builds the current reader edition. The scheduled runner repeats dependency verification after source changes; it does not invent missing physical tests or certify engineering completion.

Open the current Workbench, review calculations, CAD and videos, read the implementation and open work, or download the primary reader book. Wheel/rail tools are retained behind explicit `legacy` commands.

OpenSourceRail v0.5 primary infrastructure adapter

engineering/ferrite_maglev/README.md

The selected architecture is ferrite PM-biased active maglev, with onboard windings/electronics, passive guidebeams, local manufacturing and Rust controls on off-the-shelf SBCs. The migration decision separates the selected architecture from engineering acceptance. Accepted engineering and physical releases remain zero.

```
./osr                        # primary architecture and migration status
./osr build                  # resume dependency build of stale kits, replays, cities, report and book
./osr design run             # regenerate bounded calculations and review images
./osr design check           # independently regenerate and compare publication outputs
./osr design verify          # verify current publication source/file checksums
./osr design release-check   # exits 3 while acceptance remains open
./osr city baghdad           # separate provisional maglev planning package
./osr sim baghdad --cycles 2  # representative-train offline SOC/charging replay
cargo test -p osr-maglev-control --all-targets
cargo run -p osr-maglev-control --release --example multimodule_replay
cargo build -p osr-maglev-control --release --example coupled_bridge
tools/automation/osr-python -m engineering.ferrite_maglev.coupled --duration-s 3
```

`./osr capabilities` now publishes the primary maglev source/command directory; `./osr catalogue index|plan|run` indexes or rebuilds only primary city packages. Use `./osr catalogue plan --city baghdad` before a selected rebuild. The default `./osr readiness` reports current-source maglev gates, and `./osr test` runs the primary Rust, Python, configuration, report and browser checks. Retained platform checks and city experiments require `test legacy`, `catalogue legacy`, `capabilities legacy`, `readiness legacy` or `example-city legacy` explicitly.

The primary development-host profile contains the actual Rust replay entrypoints and no physical bindings. The five old host images remain explicitly scoped comparators; hot-axle and points services are not selected into the primary development host. Shared traction, braking, odometry, protection and passenger interfaces still require revalidation.

Open the generated report. The primary Workbench, public overview and reader book now use this architecture by default. `./osr workbench legacy`, `./osr book legacy`, `./osr city legacy` and `./osr design legacy` retain explicitly selected comparator workflows.

`./osr build` and the installed 15-minute timer use the same restartable dependency runner. It verifies native interface/bay exports, the beam-only kit, linked mechanical/civil/Rust evidence, the two-way replay, all city packages, the report, the overview and the book in dependency order. Only stale implemented outputs are rebuilt. A failed execution or rejected result holds downstream stages; a restart verifies existing evidence before resuming. An interprocess lock excludes simultaneous timer/manual dependency builds. Installing the primary timer retires this checkout's old automatic assembly and wheel/rail design/publication services; shared ERP feedback and telemetry are left under their existing management. Local progress and failure details are retained in `build/maglev-workflow/latest.json`. This is a finite build of implemented evaluators, not an autonomous engineering-completion or certification engine.

The scalar study retains 4 mass variants, 12 cost configurations, 2 additional local-manufacture budget targets, 16 normal/hot load-duty cases, 3 additional no-regen/90 km/h cases and 27 guidebeam geometry/span combinations. The 5% hot uphill steady power envelope is a separate calculation. These use declared assumptions; actual magnetic force/loss data remain open.

Implementation and closure status

Capability	Implemented scope	Evidence still needed
Transition register and explicit decisions	345 canonical identities plus 69 additional Baghdad records: 202 redesign, 17 replace, 151 retain shared with interface revalidation, 44 retire wheel-specific; all 63 old packages and 72 requirements map to 11 primary packages	Accepted exact-configuration outputs for every allocated item; legacy geometry/tests do not transfer
Native geometry	Default 624-solid partial 17 m frame/24-cartridge lift interface, plus a two-direction 25 m civil bay; actual FreeCAD solid export, FCStd/STEP reopening, identities and selected interference checks	Full body/propulsion/capture definition, qualified joints/retainers, structural calculations, materials and production drawings; export receipts identify the exact checked variant
2D field solver	Variable-reluctivity finite-difference A_z equation, manufactured sine-solution second-order controls, artificial nonlinear iron law, air Maxwell contours and gap/temperature/current coupon studies	Coupon force remains unconverged under mesh refinement; actual B-H/recoil curves, 3D ends/windings, force/loss/demagnetisation maps and physical rig correlation
Rust module array	24-module force allocation, virtual plant/replay, I/O contracts and support-aware fault logic	Actual force/current/thermal plant, isolated physical I/O, independent protection/power domains and measured SBC deadlines
Two-way beam/control replay	Rust module forces load a 50-element, six-vertical-mode beam; resulting local displacement changes subsequent controller gaps; three-second virtual demonstrator and independent static beam controls	Flexible body, torsion/lateral/soil/joint/moving-load dynamics, qualified magnetic actuator maps, physical timing and instrumented stability/ride/fault evidence
City compiler and catalogue index	266 cities / 1,565 lines with retained geography/demand, matched AW2/AW3 formations, recalculated duty/fleet/charging/depot quantities and partial low/base/high costs	Located terminal loops/ring closure, coupling/platform/egress fit, real route/charging/berth/dispatch/recovery duty and complete programme cost

The native interface is a partial material model. Its recessed-retainer variant models 8 mm nominal fastener-head clearance; the earlier 576-solid flat-retainer variant has only 2 mm. Ground ferrite edge recesses, low-profile fasteners, retention after crack/debond, force area and the complete manufacturing/control/ thermal gap budget remain unqualified. Neither nominal clearance nor successful CAD reopening establishes physical fit or a released vehicle.

The two-way replay uses a stationary rigid car and synthetic instantaneous force channels. It exchanges forces and beam displacements in both directions, but its 1 ms step is virtual time. It does not demonstrate physical I/O timing or a validated six-degree-of-freedom vehicle/guideway system. PDE residuals and the manufactured equation's refinement order likewise do not establish coupon force convergence or actual lift capacity.

City packages remain separate from the retained wheel/rail city outputs. Formation preserves both passenger loading cases; old train counts, turnouts and cost totals are not adopted. Fixed terminal loops remain unlocated allowances, charging and berth layouts require validation, and complete city programme costs remain unknown. Partial fleet/ordinary-elevated allowances are not project bids or achieved savings. The Baghdad package shows these assumptions and open gates for the six-car comparison.

Manufacturing and acceptance

The register includes 23 planned EBOM objects, 437 industrial obligations, ERP/lifecycle and observation-only FUXA interface plans, unanswered supplier questions, and G0–G5 procedures with traveller prerequisites. Its 27 unresolved quantitative thresholds must be derived and approved before testing. No material certificate, priced origin-traced MBOM, actual equipment/process qualification, installed serial, physical result or acceptance signature is invented. Supplier questions distinguish UK manufacture from UK distribution/supply; local assembly does not establish input origin or local value-added percentage.

`workbook.py` imports literal cells with source addresses and a workbook hash; it does not execute spreadsheet code or trust cached formula outputs. `model.py` recalculates the supplied budgets and physical screens. `delivery.py` binds source, input, runtime and output hashes. Checksum verification establishes integrity; `check` repeats its declared calculations. Native exports and physical tests require their own configuration-specific receipts and evidence.

Old physical generators remain accessible through explicit `legacy` launcher paths. Their city/CAD/book outputs are comparator evidence; new models and qualified physical tests must establish the maglev design.

Predominantly elevated and simpler by design

The primary policy now targets 99% elevated mainline coverage and proposes all ordinary running sections on viaducts. Existing at-grade planning lengths enter the new ordinary viaduct allowance; bridges and other special crossings retain separate quantities and unpriced scope. Existing routes are retained geography, not newly approved straight alignments. Rings and terminal loops still curve. Housing demolition, road conflicts, wildlife effects, utilities, station/egress access, temporary works and pier/foundation placement require their own evidence. An elevated line does not establish zero impact or eliminate air-rights needs.

optimisation.py runs a bounded joint search controlled by optimisation.toml. Section dimensions, span/support count, transport, lift count/mass and candidate vehicle tare enter the same comparison. The Pareto objectives include indexed civil cost, counted concrete mass, bare lift mass, car tare, train allowance and the number of unqualified vehicle target changes. Gross uncracked stiffness and bare lifting-mass screens are numerical filters; relative magnetic gap, strength, actual crane capacity and site acceptance stay open.

Costs are indexed from the supplied workbook: structural volume, support count, transport mass and handling affect their declared rows. Passive rails, walkway brackets, temporary works, installation, night work, QA, relocation and traffic receive no automatic deletion credit. A cheaper fully manufactured and erected system needs evidence for actual unit rates, connections, tooling, yield, production throughput, access and maintenance. Complete city totals stay unknown.

Vehicle targets retain equal passenger loads, storage, ferrite, capture/braking and growth reserve. Reduced frame/equipment mass does not automatically reduce procurement cost. The existing USD90,000/USD60,000 magnetic hardware targets are unquoted budget sensitivities; removed wheel/gear costs are credited only once. Doors, HVAC fans/pumps as required, cooling, sensors, support drives and independent protection remain. Imported-input cost and local value-added remain unknown until origin-traced records exist; zero normal-running-gear moving parts is an objective, not a count of every mechanism on the train.

Source context: FTA aerial fixed-guideway planning identifies ROW, adjacent-building, utility and station-access considerations. FHWA prefabrication guidance supports offsite assembly and reduced onsite disruption; its decision framework retains connections, erection equipment and local production capability as feasibility issues. These references do not validate OSR pricing or railway acceptance.

The initial expanded grid also explores 80 mm walls and deeper narrow sections. These require separate reinforcement/cover, prestress, local-load, durability, torsion and production assessment. The new screens count the retained wide iron faces from native material quantities (about 300 kg/m per direction), rather than substituting the older 100 kg/m allowance. This does not close bracket/anchor mass. Analytical alternatives are separate configurations from the retained native CAD and the earlier flexible-beam replay; those receipts do not validate the new minima.

Integrated guidebeam and onboard evacuation target

The new target is one integrated structural guidebeam per direction without a continuous external walkway. The 54-part beam-only bay removes 20 walkway cassettes and 14,940 kg of modelled material from the retained 25 m reference. Iron reaction faces are necessary for the current attraction circuit; plain concrete is not its magnetic reaction material. Supports, foundations, joints, mounts, drainage and emergency/maintenance access still need engineering.

evacuation.py and evacuation.toml define a preferred vertical controlled-descent chute concept (internal spiral or cell variants), an inclined-slide comparator and counterexamples. Four chute-height envelopes (8/12/20/30 m) assume a 1.2 m diameter, 1 m stand-off and ±0.5 m lateral displacement. These are geometric sensitivities, not supplier dimensions, a wind-response model or approved operating heights. An illustrative 8 m height at 30 degrees requires a 16 m straight slide, 13.86 m horizontal run and additional landing clearance. Live roads, roofs, water, obstructed land and insufficient clearance reject that landing path. Safe geometry alone stays on hold: fire/smoke, adjacent trains, wind/heat/dust, support immobilisation, disabled and assisted evacuation, crew/responder access, inflation/reliability and the applicable authority's acceptance are unproved. No real deployment is commanded.

For a non-fire breakdown with a tenable train, the initial strategy is recovery to a station or qualified rescue transfer. Emergency chutes/slides are not assumed to be the universal sole escape route. Historical FTA guidance and aviation/marine analogies inform research; their test criteria or equipment approvals do not transfer to this railway. The current full rail standard and deployment-specific acceptance basis require independent review. CIV-004 already retains the reference walkway until an approved egress design replaces it; that acceptance remains open.

Another 432 no-walkway research cases are held outside reference champion and city-price selection. Gross avoided civil allowances are reported separately from net system savings. A provisional three doors per side would mean six evacuation packs per car; installed mass, mounting, lost cabin space, inspection, repacking/replacement and lifetime costs require actual supplier data. The 25/50/ 100 kg pack values are mass sensitivities, not selected equipment. British supply and service leads are distinct from confirmed British manufacture.

Run `./osr design evacuation-native` to export/reopen the candidate native solids and render their retained geometry. The primary report links the CAD, counterexamples, quantities and clearance diagram. No evacuation, manufacturing or physical release is claimed.

A further geometric counterexample tests an inward-facing door against the retained adjacent beam: the straight slide centreline intersects the adjacent beam's web. An outward path clear of those modelled boxes still has no approved landing or evacuation. Door equipment count therefore cannot be used as the count of simultaneously usable exits. Both cases are retained in the structured study.

The chute research compares Axel Thoms (vertical internal spiral), Survitec MarinArk 2 (enclosed helical marine path) and VIKING Basic EVAC (offshore chute, published installation heights up to 35 m). These applications have different supports, stabilisation and receiving areas from an onboard train system. Safelincs lists a 4–120 m building spiral and up to 15 persons/minute: UK supply of a German-made product, not confirmed British manufacture or railway qualification. The manufacturer states that each installation is designed for its location.

At that unverified building flow claim, 960 passengers and 18 usable chutes give an ideal queue bound of 243 seconds including an assumed 30-second deployment; with nine usable chutes it is 457 seconds. These are optimistic arithmetic bounds, not evacuation simulations: cabin distribution, descent, assistance, entry/exit queues and blocked outlets still require evaluation. No rail time limit is adopted. A vertical swept-envelope check also retains adjacent-beam clashes where assumed sway reaches the other guidebeam. Ground anchors, variable length at arbitrary stop positions, outlet control, local fire/smoke, packed volume and door/body reaction loads remain design work; no civil or lifecycle saving is accepted yet.

Implementation roadmap

This roadmap reconciles the initial v0.5 review with the current elevated, integrated-beam and spiral-chute direction. The older request for mandatory walkways is a comparator; an approved alternative is still required before removal. Architecture objectives are implemented; complete design and manufacture remain open.

Workstream	Current executable comparison	Required next evidence
Elevated, direct alignments	All ordinary retained sections proposed elevated; original class lengths and conversion budgets retained; radial path length / first-to-last chord reported from cell geometry	Located alternatives, standard straight-span fraction, horizontal/vertical radii, buildings/utilities, station walking catchments and interchange access; rings need their own defined corridor endpoints
Complete installed viaduct	Candidate-specific indexed ordinary allowances, counted concrete, bare lifts, support counts; separate lowest-allowance/lowest-counted-mass observations and Pareto set	Actual foundations/ground, prestress/reinforcement, joints, lifting accessories/crane radius, tolerances, interfaces, full installed quantities and supplier prices
Production and erection	Quantities and indexed transport/handling sensitivities	A maglev-specific finite-resource factory, curing, connection and erection schedule; fastest-construction candidate remains null until this exists
Complete rolling stock	Matched payload/storage/reserve/ferrite, mass-allocation alternatives, fixed running-gear objective, required passenger/protection equipment retained	Part-level EBOM/MBOM, carriers/interfaces, unique and moving assembly counts, wear tasks, measured module replacement and failed-train recovery times
Vehicle/civil/energy feedback	Candidate AW3 mass changes beam demand and the same hot, no-regeneration, straight/level duty/charger screen	Qualified force/loss/thermal maps, candidate-specific actuator layout/control and coupled dynamics; bidirectional complete-system optimisation is not yet implemented
Local manufacture and cost	Workbook-indexed allowances; removed conventional scope credited once; local-integration budget targets separate	Origin-traced supplier-priced MBOM, factory-gate scope, complete installed cost, labour, tooling/yield, specialist dependencies and lifecycle repair/renewal costs
Beam-only evacuation	Conditional CAD, slide/chute clearance and unsafe-landing counterexamples; no-walkway cases excluded from reference promotion	Selected device/mount/entry design, additional mass, every-stop landing/alternative, descent/flow, accessibility, wind/fire/rescue, cost and independent acceptance

Every search candidate has a `system_scorecard` in the complete retained `optimisation-study.json.gz`. Values identify their units and arithmetic scope; missing quantities are null. Cost and mass extrema refer to indexed ordinary allowances and counted concrete, not complete installed price or mass. All no-walkway scorecards hold energy/charging because extra chute mass is unknown. The profile-level baseline duty remains a comparator. Supplier-priced cost, imported value, labour and repair times are not inferred from a lighter mass target. The separate legacy civil schedule is not promoted into a maglev winner.

The manufacturer's special-constructions page explicitly lists monorail and gondola applications. Its stationary page gives a custom application range from roughly 2 m to over 110 m; this is distinct from the UK supplier's 4–120 m listing. Neither is an accepted OSR operating envelope. The mobile system has zippered exits at different heights. Their spacing, selection, outlet clearance and train-specific stabilisation are not specified by these pages. A named transport installation and its approval have not been independently verified here.

The retained study now contains seven open installation requirements: protected cassette, enclosed controlled descent, route height coverage, structural entry and anchors, ground outlet/dispersal or alternative, complete lifecycle scope, and driverless deployment/assistance. It does not assume a trained attendant is aboard. A hypothetical device with exits at 8/12/20/30 m fails the sampled 10 m height: covering minimum and maximum height alone does not cover the intervening terrain. Even exact sampled height matches remain held for usable outlet, safe receiving area, wind, assistance and every-stop coverage. No zippered-port layout or mounting design is selected. As a further geometric comparator, a 20 m inclined slide at 35 degrees projects about 28.56 m horizontally before landing.

Linked mechanical, civil and software increment

`./osr design system` now generates and reopens two native assemblies, renders images/assembly visibility videos, executes Rust egress scenarios and a separate native-derived beam/control replay, and publishes a source-bound packet at `system-development`. This is a linked research increment, not a complete vehicle or accepted railway.

The doorway demonstrator has 20 physical tube/plate/shell/deck parts, eight anchor-plate holes, an empty cassette and a 1.34 × 2 m clear opening. Four outlet stand-offs (0.8/1.2/1.6/2 m) expose clearance versus partial material mass. The illustrative 2 m version weighs about 221 kg before fabric, attachment bolts, hinges/locks, full guards or the deployment mechanism. Its extended envelope cannot be adopted as the travel configuration. A fold/stow/deploy mechanism, vehicle-body attachment load path and full supplier-exact module remain open. The partial material CSV gives dimensions and quantities; it is not a released MBOM.

The civil ledger explores 576 combinations of cap width, column dimensions, foundation envelope, additional equipment mass and load stage. It transfers two adjacent half-spans per direction into an interior support, retaining native beam/rail material and support self-weight. Symmetric/asymmetric traffic and one-beam erection expose eccentric moments and nominal column/contact demands. A negative linear footing pressure invalidates the full-contact assumption; it does not select a pile foundation or establish soil capacity. Smaller geometry has no automatic cost or acceptance credit. The 800 reference walkway unit placements/km are a separate-erection quantity, not crane cycles or days saved.

`egress::evaluate` in `osr-maglev-control` is a pure, allocation-free offline state machine with candidate/door context, monotonic/fresh inputs, latched faults, secured stop and assistance/landing/height/clearance prerequisites. Completion retains a stop until fresh, secured, positively stowed reset. Calculated hypothetical clearance and discrete-height results feed four actual Rust runs; inward clash, missing 10 m outlet and road landing prevent study deployment. Device readiness/security/assistance remain synthetic facts. No physical adapter is supplied. A test composes the egress stop with existing support control and checks that valid support continues while traction is inhibited.

The additional two-way replay recovers beam line mass from retained native CAD volumes (about 1,950 kg/m), then uses it with analytical box inertia in the actual Rust/flexible-beam demonstrator. It adds the modelled entry hardware for six hypothetical door units per car to the stationary 27,750 kg baseline in both the Rust gravity trim and rigid-car plant, retaining assumed inertias and synthetic actuator maps. Complete chute mass, centre of mass and inertias remain open. Treating the illustrated frame as added hardware is a gross sensitivity; a net MBOM must resolve any overlap with the existing body structure and emergency allocations. It does not qualify the entry frame, support variants, moving trains or full vehicle dynamics.

Baghdad reference configuration — revision 2

The typed configuration and dimension assumptions fix Baghdad at six 17 m bodies. The station study adds assumed 0.5 m inter-car gaps and 2.5 m end allowances: 104.5 m assumed train length and 109.5 m platform. These allowances remain unresolved engineering inputs. The 2.96 m magnetic arms determine beam spacing. A 0.255 m cabin-to-platform gap requires a qualified bridge plate. The centred lift has a native floor opening and zero nominal threshold step.

Retained native station, quantities and video include partial floor frames for six cars, rather than a complete vehicle. Pier-height studies at 8/12/20/30 m propagate column mass and gravity demand; the corresponding assumed door heights are 11.1/15.1/23.1/33.1 m. Chute height screens alone do not qualify these structures or outlets. Emergency-energy accounting includes stop, deployment, passenger flow and rescue; unknown device power prevents a complete reserve claim.

The common requirements register combines 72 handoff and seven evacuation requirements. `CIV-004` remains applicable until an approved route-specific alternative exists. No requirement is superseded by a study pass.

osr-maglev-control

crates/osr-maglev-control/README.md

Portable v0.5 maglev study core, using Rust `no_std`, no heap and no external dependencies. It targets simulation and off-the-shelf SBC replay. Every computed output has `physical_output_authorised = false`; the I/O boundary records study outputs and supplies no current, gate-drive or PWM operation.

The existing `evaluate(previous, inputs, now_ns, params)` API remains a single-module integer current hypothesis. Its polarity, damping, envelope, time/replay, stop/reset and support-power tests remain compatible. Increasing an upward-attraction gap asks for increasing support. The controller cannot be reused unchanged for shaped-pole passive vertical support.

24-module force and moment study

`array::evaluate_array` adds coupled heave, roll, pitch, lateral displacement and yaw feedback and `array::allocate` distributes requested forces and moments over 24 stations. This is one rigid car, not a three-car coupled train or a validated six-degree-of-freedom suspension. Longitudinal force has no actuator in this model; a nonzero longitudinal demand appears explicitly as unachieved residual. Propulsion, braking, roll/yaw cross-coupling from real traction, flexible car/beam modes, nonlinear finite-angle dynamics and contact dynamics remain open.

Coordinates use x forward, y left and z up, with the assumed car centre of mass as origin. Station positions and gaps are metres, force is newtons, moments are newton-metres, small angles are radians and timestamps are monotonic nanoseconds. Upward force is positive attraction. Lateral force is positive left. Each moment is the sum of `position cross force`. The explicit seed geometry is 12 paired stations at x = -5.5, -4.5, …, 5.5 m and y = ±0.88 m, z = 0 m; this matches the candidate 12 m magnetic strips. Actual centre of mass, attachment heights, lever arms, tolerances and dynamic mass properties need physical closure.

Each station has a bounded upward channel and a signed lateral channel. The illustrative limits are 0–20 kN upward, ±4 kN lateral and 2 MN/s slew per channel. These numbers are hypothetical force envelopes, not measured magnetic maps, current/voltage/thermal limits or proved simultaneous lift/guidance capability. A real PM-biased module can retain force after drive loss: a failed station takes an explicit specified residual force. Zero in the example is an ideal failure assumption. Unknown failed-channel forces require a separate bounded uncertainty and protection model before hardware use.

The allocator uses at most 256 bounded coordinate sweeps to minimise squared force residuals and moment residuals scaled by a declared 6 m lever arm. It reports the independently reconstructed resultant, residual wrench, constraint limits and feasibility tolerances. A solver iteration count establishes no physical deadline. Infeasible requests remain visible; no six-axis success is inferred from solver completion. Slew limits apply against a prior force frame; without one the first study allocation has no force history and applies only static limits. This is not a physical startup or touchdown procedure.

The mode estimator fits a small-angle rigid plane to 24 dual-gap samples and fits lateral displacement/yaw to 24 dual lateral samples. Every individual gap and lateral sample is checked before averaging; disagreement, nonfinite values and degenerate geometry fail. This fit is not a flexible-body observer. Sensor channels share one supplied acquisition timestamp; per-channel skew, physical independence, common-cause failure and calibration are not established. Derivatives use acquisition time. Host time/sequence checks reject stale, future, replayed, regressing and late frames.

A stop request preserves fresh bounded support. A station failure, active bound or infeasible allocation latches a traction stop while available ideal support remains active. Missing valid sensing or power requests external local fallback, never frozen study forces. A prior fault inhibits traction until a fresh healthy frame, reported stopped/secured conditions and reset coincide. Capture, fallback, independent protection energy and secure-condition verification need separate engineering; they are not supplied by these Boolean reports.

Reproducible offline checks

Run:

```
cargo test -p osr-maglev-control
cargo clippy -p osr-maglev-control --all-targets -- -D warnings
cargo run -p osr-maglev-control --example sbc_replay
cargo run -p osr-maglev-control --example multimodule_replay
```

The array replay integrates ten seconds of virtual 1 ms frames, prints CSV at 500 ms intervals and asserts feasibility and a final small five-mode error. Its 27,750 kg AW2 mass is the profile's 18,750 kg matched tare plus 120 × 75 kg passengers. Gains, diagonal inertias, force limits and instantaneous actuator response are synthetic. It starts with coupled displacement/tilt, commands a stop at 0.5 s and loses station 0 at 1 s under a specified zero-residual failure hypothesis. It keeps fresh support and the stop latch active. This trajectory is a numerical example, not a closed-loop stability or ride-quality

proof.

`ideal::IdealPlant` integrates a constant wrench exactly within each small-angle step. Tests independently check uniform equilibrium (272,227.5 N weight), moment signs, force/moment reconstruction, five-component demand, failed-force redistribution, capacity and slew infeasibility, all-failed residuals, sensor and time rejection, support-preserving stops and secure reset. Mechanical interface power agrees with resultant wrench power, and one-step work agrees with change in kinetic plus gravitational energy. These conservation checks validate the ideal mechanics arithmetic, not actual magnet energy or electrical losses.

`io::StudyIo`, `host_tick` and `ReplayIo` define an acquisition/record boundary without owning an OS clock or hardware. Recording/acquisition failures return to the host; the host must engineer its local fallback. Repeated single-frame stub reads are rejected as replay. No scheduler, GPIO, ADC or network transport is validated, and this example measures neither SBC timing nor I/O latency.

Physical gates remain open: calibrated nonlinear magnetic force/loss maps over gap/current/temperature and shared flux, identified electrical/thermal and mechanical dynamics, actuator failure residuals, independent sensors/protection, measured end-to-end target timing, multi-module rigs, flexible guideway and loaded vehicle tests, braking/capture and train/route acceptance. No operational or manufacturing authority is inherited from the numerical study.

Flexible guideway numerical demonstrator

`coupled_bridge` connects this exact array core to the separately labelled Python beam demonstrator in `engineering/ferrite_maglev/coupled.py`. From the repository root, run:

```
cargo build -p osr-maglev-control --release --example coupled_bridge
.venv/bin/python -m engineering.ferrite_maglev.coupled --duration-s 3
.venv/bin/python -m pytest -q engineering/analysis/tests/test_ferrite_maglev_coupled.py
```

The Python host sends all 24 local relative gaps each virtual millisecond. Rust returns bounded support forces, which load a 25 m simply supported Euler–Bernoulli beam; its local displacement feeds the next sensor frame. This is two-way offline coupling, with no physical output. The beam assumes E = 30 GPa, I = 0.193624 m⁴, line mass 1750 kg/m and 2% modal damping, retaining six vertical modes of a 50-element consistent-mass model. Independent point-load closed forms check static response; analytical pinned-beam frequencies and virtual-work/undamped-energy checks control the numerical implementation.

The model can show global beam displacement exceeding 8 mm while the car follows that displacement and the relative gaps remain inside the example's study envelope. This establishes neither a gap requirement nor structural acceptance. It omits dead-load camber, torsion/corbel flexibility, moving/train loads, flexible car structure, soil, joints, fatigue, real magnetic/electrical dynamics and independent protection. The line protocol accepts at most 20,000 frames, 8 KiB per frame, and measures no hardware timing. The three-mode car motion and vertical beam coupling do not provide a six-degree-of-freedom stability proof.

Offline evacuation sequencing

`egress::evaluate` adds allocation-free doorway sequencing with an exact declared candidate/door context, fresh monotonic frames, explicit readiness/assistance prerequisites, latched contradiction/context/deadline faults, and secured/stowed reset. It reports study traction inhibition and continued-support obligations; it supplies no door, restraint, chute or support output adapter. Input assertions are unauthenticated study facts, not verified independent sensor or safety evidence.

`cargo run -p osr-maglev-control --release --example egress_replay -- REVISION PATH_CLEAR HEIGHT_COVERED` prints a bounded nine-frame synthetic sequence. The linked Python study supplies its candidate tag and calculated hypothetical path/height gates, and retains negative controls. Loss of prerequisites after deployment begins holds the sequence; no automatic retraction is commanded. The egress/support composition test preserves fresh support control while inhibiting traction. Physical timing, independent protection, accessible passenger evacuation and railway approval remain open.

RFC 0035 — v0.5 primary ferrite-maglev infrastructure

docs/rfcs/0035-v05-primary-ferrite-maglev.md

Status: selected development architecture; engineering acceptance remains open. Decision date: 20 October 2026. Supersedes v0.4 wheel/rail physical defaults for new development.

The user selected a broadened ferrite permanent-magnet maglev system as the primary OpenSourceRail infrastructure. The machine-readable decision is the infrastructure policy. Its input is revision B of the maglev profile. The supplied revision-A specification and cost workbook remain the reference handoff. Statements in specification A that describe the TOML as an unimplemented scaffold describe its original handoff; the dedicated v0.5 adapter now accepts revision B. The wheel/rail generators still do not accept it.

The primary design combines fixed onboard ferrite modules, PM-biased active upward attraction, separate active guidance, onboard windings and electronics, passive guidebeams, battery/station charging and switch-free mainline topology. Shaped-pole vertical restoring suspension and linear induction propulsion remain alternatives requiring their own model adapters. Three 17 m cars, matched 675 kWh gross storage and unchanged AW2/AW3 payloads are initial study values. Baghdad's six-car comparator cannot be replaced one for one by this formation.

The system objective is locally owned design, fabrication, integration and repair. Minimise imported finished running-gear packages; develop retained magnet cartridges, carriers, winding/impregnation, module assembly/calibration, guidebeams and replaceable passive reaction surfaces locally. Optimise complete manufactured and erected function: foundations, supports, beam, magnetic interfaces, evacuation, services, transport, tooling and construction stages. Compare cost, installed mass, maximum lift, labour, throughput, alignment, maintainability and external payment obligations together.

Rust implements the portable control algorithms and SBC host integration. `osr-maglev-control` now includes an allocation-free `no_std` core, a 24-module force allocator, virtual plant/replays, I/O contracts and a support-aware traction-permission/fault policy. Fresh support control continues during a commanded stop. Invalid sensing requests a separately engineered local fallback; holding an old PWM value is not a fallback controller. Saturation inhibits traction. Fault reset requires fresh healthy inputs and reported stopped/secured conditions. Physical actuation remains disabled. Synthetic gains, ideal force channels and logical protection do not establish the actual magnetic plant, independent hardware channels or measured SBC deadlines.

The two-way numerical demonstrator connects the Rust 24-module controller to a 50-element, six-vertical-mode flexible beam. Returned support forces load the beam; local beam displacement changes the next controller gap readings. A three-second virtual replay and independent static beam checks are implemented. The car is rigid and stationary; torsion, lateral/soil/joint/moving-load dynamics, actual force/current/thermal maps and physical timing remain open. This subset does not close full six-degree-of- freedom stability, ride or failure acceptance.

Off-the-shelf SBCs are the preferred hosts. Host choice does not remove the cost of power semiconductors, isolated sensing, current control, switching I/O, protection energy, cooling and physical qualification. Real-time scheduling and local control must be demonstrated on the selected board with production I/O. Linux PREEMPT_RT reduces scheduling latency; it does not supply a measured application deadline. The published coupled motor/levitation study supports the research architecture, with no transfer of its results to this car.

Cost model changes retain the supplied low/base/high allowances, including the exact conventional removal scope, replacement support/brakes and separate NRE. Two additional $90k/$60k magnetic-bundle targets retain all other base costs. They are optimisation hypotheses. Localisation is costed local value added, with imported inputs retaining their origin. No percentage is asserted until an origin-traced priced MBOM is available. No catenary-removal credit or citywide saving is inherited. Civil component quantities and gross stiffness screens do not establish strength, gap control, support savings or a finished shipment mass.

Implemented development scope

The transition register preserves 345 canonical identities plus 69 additional Baghdad records. Its 414 explicit dispositions cover 202 redesigns, 17 replacements, 151 shared items requiring interface revalidation and 44 wheel-specific retirements. It maps all 63 legacy work packages and 72 ferrite requirements to the 11 primary packages. Every allocation carries deliverables and open evidence actions; inventory coverage is not engineering completion or acceptance. Wheel retirement keeps replacement support, capture, braking, parking and recovery obligations visible.

The geometry adapter now defines a partial 624-solid 17 m frame/24-cartridge lift interface and a two-direction 25 m civil bay. The native adapter constructs actual FreeCAD solids and exports FCStd/STEP with reopening and selected

identity/interference checks. The exact export receipt identifies the checked variant. The default recessed-retainer geometry models an 8 mm nominal fastener-head gap; its earlier 576-solid flat variant has 2 mm. Ferrite edge grinding, retention, fasteners, full error budget, complete vehicle equipment, structural joints, reinforcement and foundations remain engineering/manufacturing holds.

The field backend solves a 2D variable-reluctivity equation and verifies a manufactured sine solution with second-order refinement controls. An artificial nonlinear iron law, air Maxwell contours and gap/temperature/current coupon sensitivities are available. Coupon force is not mesh-converged; the equation control does not establish magnetic force accuracy. Actual material grades, 3D geometry/ends, speed-dependent losses, demagnetisation and vehicle lift capacity require qualified models and rig data.

The city compiler has generated separate provisional packages for 266 cities and 1,565 lines. It retains geography/demand while recomputing passenger-matched formation, duty/fleet/charging/depot quantities and partial cost scenarios. Ordinary city and simulation launchers now invoke this compiler and an offline representative- train SOC replay. Located loops/ring closure, actual route and charging limits, platforms, berth conflicts, depot access/recovery and complete programme costs remain open. Baghdad's old six-car fleet count and cost total do not transfer.

The register also supplies 23 planned EBOM objects, 437 industrial obligations, ERP/lifecycle and observation-only FUXA contracts, traveller prerequisites and supplier questions separating manufacturing origin from supply/distribution location. Its G0–G5 protocols retain 27 unresolved acceptance thresholds; limits must be approved before tests and physical evidence must be authenticated. No supplied material origin, quote, local-content percentage or physical result is asserted. Accepted engineering/physical releases: zero.

Migration contract

Surface	v0.5 behaviour	Remaining work
Version and policy	`VERSION` 0.5; Rust/frontend 0.5.0; primary family selected	Configuration-specific deployment acceptance
`./osr`, `./osr design`, `./osr build`	Primary status, source-bound study publication and Rust examples	Complete engineered CAD and material/force/loss-qualified solvers
`./osr design legacy …`, `./osr sim legacy …`, `./osr city legacy …`, `./osr build legacy`	Explicit wheel/rail comparator access	No acceptance transfer
Ordinary `./osr city …` and `./osr sim …`	Provisional maglev city compilation and representative-train energy/charging replay	Located topology, route/berth/charging/recovery acceptance and complete programme costs
Existing wheel/rail catalogue, CAD, videos, book and brochure	Retained v0.4 physical comparators; maglev city packages and partial native models have distinct identities	Complete maglev configuration and qualified manufacturing outputs
Platform GUI, ERP and supervision	Retained tools plus planned maglev item/lifecycle/tag contracts	Released priced MBOM, actual entity/serial records and commissioned observation bindings; no live control authority
Background design	Source-bound primary study refresh; native geometry, 2D field and two-way beam/control subsets available	Broader validated electromagnetic/structural models, target hardware and physical evidence

Progressive completion

1. Close geometry, support/module placement, magnet allocation, force directions, vertical curves, twist, tolerance budget and switched-free terminal topology.
2. Complete magnetic/coil/rail production geometry from the partial native model; converge coupon force and qualify actual material, 3D, loss, finite-end and joint/temperature models against independent and physical evidence.
3. Derive thermal/current/voltage envelopes, force redistribution, 6-DOF control, flexible-body/guideway dynamics and physically independent protection.
4. Extend provisional city compilation/replay to matched 80/90 km/h, actual hot/grade/curve/wind, full-SOC/no-regen, low-voltage, missing-charge and failed-channel duty. Validate fleets, platforms, berths, depots, charging and switch-free recovery against the accepted service and physical route.
5. Derive EBOM/MBOM, local/import origin, complete BoQ, processes, fixtures, yield, labour, QA, spare modules and lifecycle repair evidence.
6. Execute G0–G5 with predetermined acceptance limits and authenticated evidence.

The adapter executes available finite calculations and distinguishes implemented subsets from missing engineering and acceptance evidence. Source or requirements changes trigger a new result. Repeating unchanged arithmetic cannot manufacture missing field maps, measurements or accepted designs. All physical/manufacturing release flags remain false.

Native geometry review



Partial fixed magnetic interface; not a complete vehicle.

Partial fixed magnetic interface; not a complete vehicle.



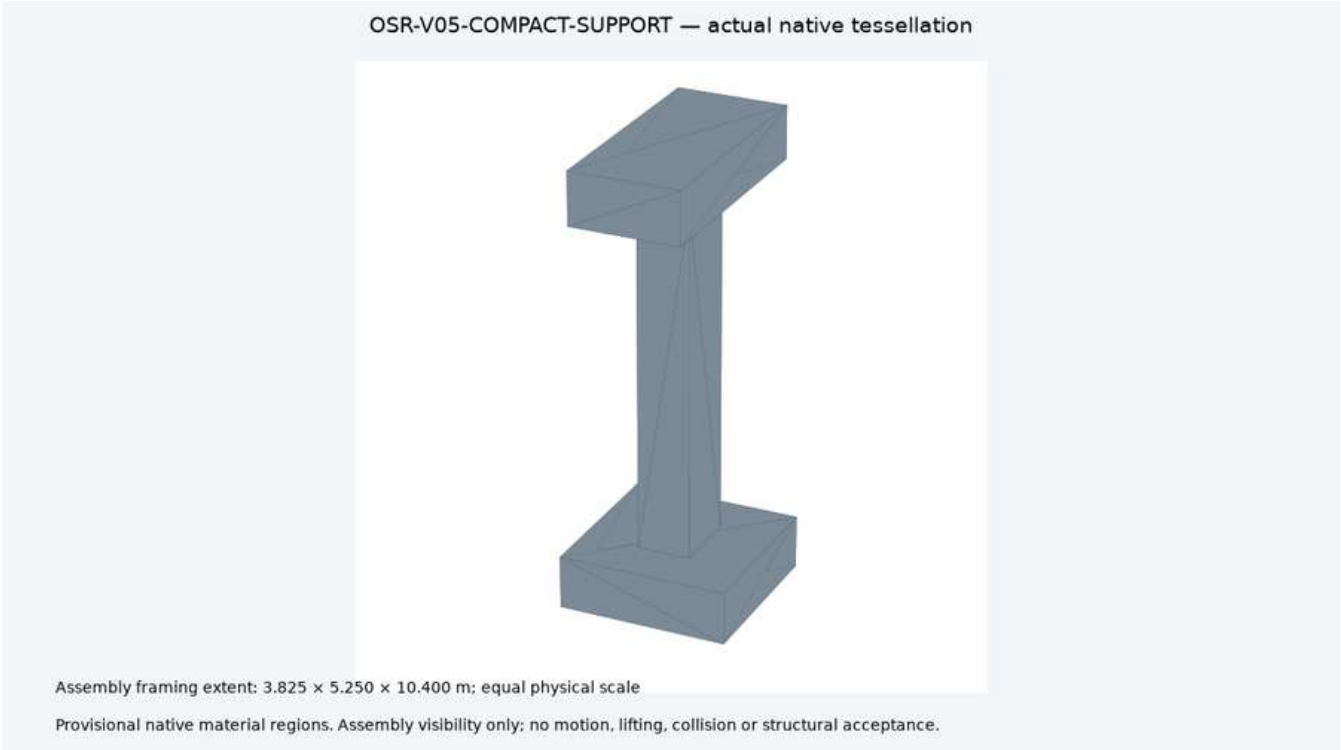
Beam-only civil variant. Supports are envelopes; evacuation acceptance remains open.

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Partial evacuation entry frame and empty cassette. No deployed or stowed mechanism is qualified.

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Compact pier and foundation envelope. Site capacity and structural acceptance remain open.

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Primary maglev city directory

Separate planning packages. Complete programme costs, local content, land and physical acceptance remain open for every city.

City	Provisional trainsets	Cars	Open gates
Benguela	115	345	12
Huambo	91	273	12
Lobito	64	192	12
Luanda	650	3900	12
Lubango	119	357	12
Malanje	28	84	12
Namibe	79	158	12
Soyo	63	126	12
Uige	17	51	12
Bukavu	118	354	12
Goma	85	255	12
Kananga	50	200	12
Kinshasa	1334	8004	12
Kisangani	64	256	12
Lubumbashi	192	768	12
Mbuji Mayi	132	528	12
Eldoret	133	399	12
Garissa	48	96	12
Kakamega	67	134	12
Kisii	50	100	12
Kisumu	79	237	12
Kitale	66	132	12
Machakos	64	128	12
Malindi	53	106	12
Meru Ke	64	128	12
Mombasa	359	1436	12
Nairobi	822	4932	12
Naivasha	86	172	12
Nakuru	84	252	12
Nyeri	57	114	12
Thika	98	294	12
Antananarivo	536	3216	12
Beira	69	207	12
Chimoio	54	162	12
Lichinga	28	56	12
Maputo	249	996	12
Nacala	77	154	12
Nampula	80	240	12
Pemba Mz	66	132	12
Quelimane	15	45	12
Tete	88	264	12
Xai Xai	40	80	12
Huye	73	146	12
Kigali	204	816	12
Rubavu	61	122	12
Mogadishu	232	928	12
Waw	45	90	12
Arusha	87	261	12
Dar Es Salaam	800	4800	12
Dodoma	80	240	12
Iringa	64	128	12
Kigoma	52	104	12
Mbeya	77	231	12
Morogoro	76	228	12
Moshi	57	114	12
Mwanza	251	1004	12
Shinyanga	90	180	12

City	Provisional trainsets	Cars	Open gates
Songea	19	38	12
Sumbawanga	35	70	12
Tabora	47	94	12
Tanga	83	249	12
Zanzibar City	83	249	12
Arua	67	134	12
Entebbe	78	156	12
Fort Portal	55	110	12
Gulu	129	387	12
Hoima	53	106	12
Jinja	92	184	12
Kampala	300	1200	12
Lira	88	176	12
Masaka	62	124	12
Mbale	64	128	12
Mbarara	89	267	12
Soroti	21	42	12
Lusaka	337	2022	12
Lyon	357	1428	12
La Paz	252	1008	12
Cuenca	101	303	12
San Salvador	294	1176	12
Agadir	120	360	12
Beni Mellal	46	92	12
Fez	191	764	12
Kenitra	105	315	12
Khouribga	39	78	12
Marrakech	322	1288	12
Meknes	82	246	12
Nador	67	134	12
Oujda	56	168	12
Safi	64	192	12
Tangier	247	988	12
Tetouan	98	294	12
El Obeid	81	243	12
Kassala	41	123	12
Khartoum	550	3300	12
Nyala	72	216	12
Omdurman	441	1764	12
Port Sudan	61	183	12
Tunis	275	1100	12
Bloemfontein	190	570	12
Durban	586	3516	12
East London Za	91	273	12
Nelspruit	57	114	12
Polokwane	83	249	12
Herat	64	192	12
Jalalabad Af	77	231	12
Kabul	466	2796	12
Kandahar	88	264	12
Mazar E Sharif	111	333	12
Barisal	108	324	12
Chittagong	519	3114	12
Comilla	150	450	12
Gazipur	311	1244	12
Khulna	255	1020	12
Mymensingh	85	255	12
Narayanganj	123	369	12
Rajshahi	112	336	12
Rangpur	70	210	12
Sylhet	81	243	12
Agra	159	636	12
Bhopal	235	940	12

City	Provisional trainsets	Cars	Open gates
Coimbatore	435	2610	12
Indore	412	2472	12
Jodhpur	176	704	12
Kanpur	378	2268	12
Lucknow	371	2226	12
Madurai	243	972	12
Meerut	154	616	12
Patna	398	1592	12
Raipur	154	616	12
Rajkot	133	532	12
Ranchi	283	1132	12
Vadodara	198	792	12
Varanasi	222	888	12
Vijayawada	206	824	12
Visakhapatnam	264	1056	12
Biratnagar	55	110	12
Kathmandu	373	1492	12
Pokhara	126	378	12
Bahawalpur	79	237	12
Faisalabad	235	1410	12
Gujranwala	207	828	12
Hyderabad Pk	296	1184	12
Karachi	689	4134	12
Larkana	50	150	12
Multan	164	656	12
Peshawar	204	816	12
Quetta	125	500	12
Rahim Yar Khan	99	297	12
Sheikhupura	58	174	12
Sialkot	83	249	12
Sukkur	73	219	12
Colombo	473	2838	12
Galle	99	297	12
Jaffna	95	285	12
Kandy	107	321	12
Phnom Penh	432	1728	12
Bandung	419	1676	12
Surabaya	469	2814	12
Vientiane	117	351	12
Mandalay	266	1064	12
Yangon	691	4146	12
Davao	309	1236	12
Ouagadougou	294	1176	12
Bafoussam	95	285	12
Bamenda	80	240	12
Bertoua	36	108	12
Douala	243	1458	12
Edea	11	22	12
Garoua	59	177	12
Kumba	62	186	12
Maroua	80	240	12
Ngaoundere	44	132	12
Yaounde	334	2004	12
Conakry	139	556	12
Bamako	413	1652	12
Niamey	233	932	12
Aba Ng	56	168	12
Benin City	230	920	12
Ibadan	259	1554	12
Ilorin	79	237	12
Jos	91	273	12
Kano	559	3354	12
Maiduguri	209	836	12

City	Provisional trainsets	Cars	Open gates
Onitsha	231	924	12
Port Harcourt	231	924	12
Uyo	63	189	12
Dakar	372	2232	12
Arish	24	48	12
Asyut	80	240	12
Beni Suef	55	165	12
Damanhur	82	246	12
Damietta	110	330	12
Fayoum	92	276	12
Hurghada	54	108	12
Ismailia	90	270	12
Kafr El Sheikh	60	120	12
Luxor	103	309	12
Mahalla	58	174	12
Mansoura Eg	75	225	12
Minya	89	267	12
Port Said	54	162	12
Qena	95	285	12
Sohag	84	252	12
Suez	89	267	12
Tanta	107	321	12
Zagazig	64	192	12
Amarah	76	228	12
Baghdad	550	3300	12
Baqubah	110	330	12
Basra	399	2394	12
Diwaniyah	88	264	12
Duhok	86	258	12
Erbil	202	808	12
Fallujah	80	240	12
Hillah	95	285	12
Karbala	208	832	12
Kirkuk	197	788	12
Kut	70	210	12
Mosul	257	1028	12
Najaf	246	984	12
Nasiriyah	79	237	12
Ramadi	85	255	12
Samawah	84	252	12
Sulaymaniyah	121	484	12
Amman	440	2640	12
Aqaba	55	110	12
Irbid	76	228	12
Zarqa	103	309	12
Beirut	214	856	12
Sidon	54	108	12
Tripoli Lb	72	216	12
Gaza City	47	141	12
Hebron	108	324	12
Nablus	109	327	12
Abha	99	297	12
Al Kharj	98	294	12
Buraidah	100	300	12
Dammam	313	1252	12
Hail	116	348	12
Hofuf	111	333	12
Jeddah	491	2946	12
Jizan	90	270	12
Khamis Mushait	148	444	12
Mecca	252	1008	12
Medina	208	832	12
Najran	88	264	12

City	Provisional trainsets	Cars	Open gates
Tabuk	96	288	12
Taif	93	279	12
Aleppo	383	1532	12
Damascus	294	1176	12
Deir Ez Zor	90	270	12
Hama	111	333	12
Homs	68	204	12
Idlib	63	126	12
Latakia	90	270	12
Raqqa	69	207	12
Tartus	58	116	12
Aden	81	243	12
Dhamar	58	116	12
Hodeidah	48	144	12
Ibb	96	288	12
Lahij	49	98	12
Mukalla	122	366	12
Sanaa	405	2430	12
Sayun	42	84	12
Taiz	89	267	12